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RESEARCH

# Describing posture on the bike: do different methods lead to similar outcomes?

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## Abstract

**Purpose** Standardising methods to calculate joint angles is essential to enable the reproducibility of movement analysis in cycling. This study compared three methods for determining lower limb posture on the bike across three positions on the saddle.

**Methods** Fourteen non-cyclists were assessed in two laboratory visits. The first involved determining their maximum aerobic capacity which was used in the second visit to define a sub-maximal cycling exercise intensity. Lower limb kinematics were obtained and angles for the hip, knee, and ankle were calculated using three methods (6 o'clock position, minimum knee angle, and the largest lower limb extension). Two-way analysis of variance with repeated measurements was utilised for the hip, knee, and ankle data to assess the main effects from the method of measuring angles (i.e. 6 o'clock position vs. minimum knee angle vs. largest lower limb extension) and from positions on the saddle (i.e. anterior vs. reference vs posterior), followed by post-hoc analyses.

**Results** The angles obtained at the 6 o'clock position were larger than those at the minimum knee angle and the largest lower limb extension for the hip, knee, and ankle joints ($p < 0.01$). Knee flexion was greater at the anterior position than the posterior and the reference, with larger angles for the reference than the posterior ($p < 0.01$). The ankle was more dorsiflexed at the anterior vs. posterior positions, anterior vs. reference positions, and references vs. posterior positions ($p < 0.01$).

**Conclusion** All three methods were sensitive to detect changes in saddle position but data should not be interchanged due to differences in angles between methods. These results indicate that body position on the bike should always be described using a consistent method.

**Keywords** Bicycle · Kinematics · Angles · Biomechanics · Cycling

## Introduction

The body of work in cycling biomechanics has largely concentrated in assessing implications from exercise intensity [1–3], changes in cadence [4–6], consequences from pain and/or injury [7–9], fatigue effects [10–12], and changes in posture on the bike [13–15]. In terms of body position on the bike, the research studies aim to explore strategies to improve performance [16, 17] or to reduce the incidence of non-traumatic injuries [10, 18]. This research has provided important background information but gaps exist in terms of how to describe posture on the bike.

The description of posture on the bike has been largely utilised to understand differences between road cyclists and triathletes [19] and to explore associations between movement patterns and injuries [20, 21]. This is a key component of bike fitting, which involves improving the configuration of the bike to optimise performance, increase comfort, and reduce the risk of non-traumatic injuries [21–23]. However, differences in the way bike fitting is conducted lead to discrepancies in bike setup for the same cyclist [24], which could be partially attributed to the methods used to determine posture on the bike.

Historically, bike fitting has been conducted using anthropometric measurements of cyclists, which would be

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'matched' with bike components [25]. The criticism of this method has been presented [26], moving the industry to the assessment of posture on the bike using static poses [27, 28]. The problem with static poses is the discrepancy between static angles compared to angles taken whilst cycling [29]. The recommendations were changed to support practitioners in obtaining angles from movement, posing a question about what would be the 'optimal angles' to improve outcomes from bike fitting [22, 30]. With limited research addressing this topic, the only component that has gained some clarity is the analysis of knee flexion angles to determine lower limb posture and saddle height. A systematic review reported that the knee flexion angle to optimise performance and reduce the risk of injuries should be between 30 and 43° at the bottom of the crank cycle (full extension = 0°) [31]. However, studies have used different procedures to extract the knee angles from the whole angular trace during cycling. Some studies used the 6 o'clock position method, where angles for the hip, knee, and ankle are measured when the crank is closest to the bottom dead centre [32]. Hip, knee, and ankle have also been obtained at the crank position of the minimum knee flexion angle (minimum knee angle method) [33] and at the crank position of the largest distance between the hip joint and the pedal axle [i.e. largest lower limb extension method; 34]. The assumption behind these methods is that they would provide a measure of minimum knee flexion (or maximum lower limb extension) at the bottom of the crank cycle. However, an objective comparison of these procedures has not been undertaken to ensure that angular data could be interchangeable.

One of the potential challenges in the interchangeability between the methods described before is related to changes in seat tube angle and/or saddle setback. When cyclists move their saddles backward, the minimum knee flexion happens earlier in the crank cycle, which creates a discrepancy in relation to the angle at the 6 o'clock position. In addition, the combination of changes in hip and ankle angles could potentially lead to differences in relation to the largest lower limb extension method. Addressing this question is critical to attempt standardising the analysis of cycling into consistent phases, similar to what is performed in walking and running gait [35, 36]. Moreover, this analysis could assist with more informed methods to describe posture, which is currently driven by commercial entities with no transparent research supporting the proposed methods and reference data [37].

Therefore, the aim of this study was to compare three methods for determining lower limb posture on the bike across three positions on the saddle. The hypothesis was that the 6 o'clock position method would produce different results in comparison to the minimum knee flexion angle and the largest lower limb extension methods. These differences would be more prominent when cyclists pedal at a more posterior position on the saddle.

## Materials and methods

### Experimental design and participants

This study utilised a comparative cross-sectional design to assess differences between methods to determine the posture of the lower limbs on the bike in three positions on the saddle. A sample size calculation was conducted to ascertain the required number of participants to obtain the required power to answer the research question with $\alpha = 0.05$ and the power of the test is 0.80 using G*Power statistical package [38]. We based our calculation on the intention to determine a minimum difference of 5° in knee angle, which is at the centre of the range proposed to determine body position on the saddle [i.e. 10 deg. 23, 27]. We also assumed that the within-cyclist's variability in knee angles would be 3.4° [28], resulting in an effect size of 1.47. This resulted in a sample size of six (6) participants using an ANOVA for repeated measures with within-between effects and interactions (adjusting the effect size to 0.735), which we deemed small. Therefore, we expanded our sample size to fourteen (14) participants.

Healthy male adults (non-cyclists) were recruited to avoid existing adaptations from trained cyclists to a given position on the saddle. These participants were physically active performing activities such as strength training, outdoor running, team sports, and were free from any injuries or chronic conditions affecting their ability to perform exercise. At the time of data collection, they were 25 ± 3 years of age, 175 ± 5 cm of stature, and 71 ± 7 kg of body mass. Before data collection, all cyclists signed an informed consent to participate in the study, which was approved by the University Human Ethics Committee (Project number 17684), according to the Declaration of Helsinki.

### Protocol and data collection

Data was collected across two visits to the laboratory as illustrated in Fig. 1.

During the first visit, height and body mass were measured [39]. After that, the bicycle (T-mobile, medium size, Giant, Taichung, Taiwan) saddle height was determined to ensure that the knee angle at the bottom of the stroke was between 30 and 40° of flexion. The participants then warmed up at 150 W for 10 min followed by an incremental maximal step test to exhaustion with an initial workload of 100 W and increments of 25 W/min [40] using the road bicycle attached to a cycle trainer (Computrainer Racemate, Inc., Seattle, USA). Power output was recorded throughout the incremental test along with gas exchanges

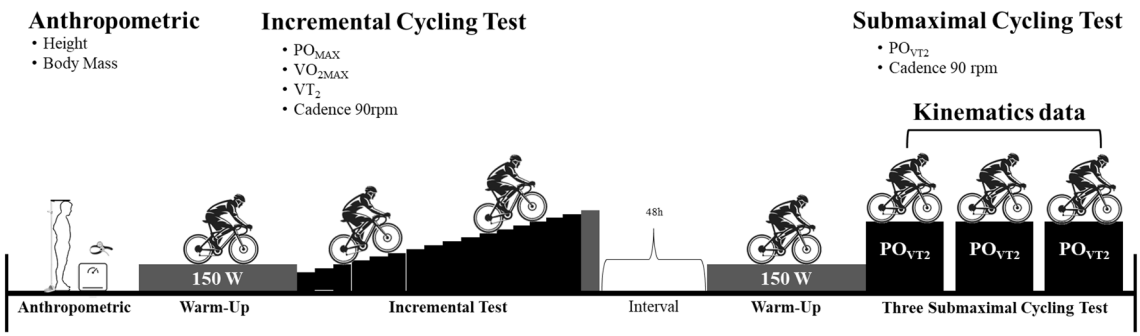


**Fig. 1** Illustration of the protocol. After anthropometrical measurements, the participants performed a warm-up followed by an incremental load test on the first visit. In the second visit, after warm-up, participants performed three bouts of sub-maximal intensity exercise, in three different saddle positions, determined by the results of the first visit

by the breath-by-breath method using an open-circuit gas exchange system (MGC CPX/D, Medical Graphics Corp., St Louis, MO, USA). Before the incremental test, the oxygen and carbon dioxide analysers were calibrated using medical grade gases that spanned air in the physiological range. Gas exchange data were analysed to define the power output at the first and second ventilatory thresholds based on the ventilatory equivalent method [41]. Pedalling cadence was controlled close to 90 ± 2 rpm using feedback from the cycle trainer head unit. The test was stopped by voluntary exhaustion or when participants were unable to maintain pedalling cadence.

After 48 h, the participants returned to the laboratory for a second visit. They warmed up for 10 min at 150 W and then cycled for 1 min with 90 rpm pedalling cadence at a power output equivalent to their second ventilatory threshold in the three positions on the saddle: reference (same position determined in the first visit), most forward, and most backward. For the forward and backward positions on the saddle, the participants were instructed to move anteriorly or posteriorly on the saddle as much as possible, without substantial changes in the hand position on the handlebars. The order of the forward and the backward saddle positions was allocated randomly. Figure 2 illustrates changes in position on the saddle.

Right lower limb kinematics were recorded for the last 20 s for each position on the saddle. As landmarks for the hip, knee, and ankle joint axes, reflective markers were placed by the same investigator on the right side of the anterior superior iliac spine, greater trochanter, lateral femoral condyle, lateral malleolus, anterior and posterior pedal stick (Fig. 3). One marker was attached to the sacrum to measure the position in relation to the bicycle frame at the different saddle positions. Two markers were taped to the bicycle frame and used as a reference for image scaling. Videos were collected from the sagittal plane using one high-speed camera (60 frames per second—fps) positioned perpendicular to the motion plane (AVT PIKE F-032, Allied Vision Technologies GmbH, Germany), using AVT ActiveCam viewer software (Allied Vision Technologies GmbH, Germany).

## Data analysis

Using DgeeMe software (Video4Coach, Denmark), video files were digitized and markers were automatically tracked for x–y coordinates over time. Kinematics were smoothed

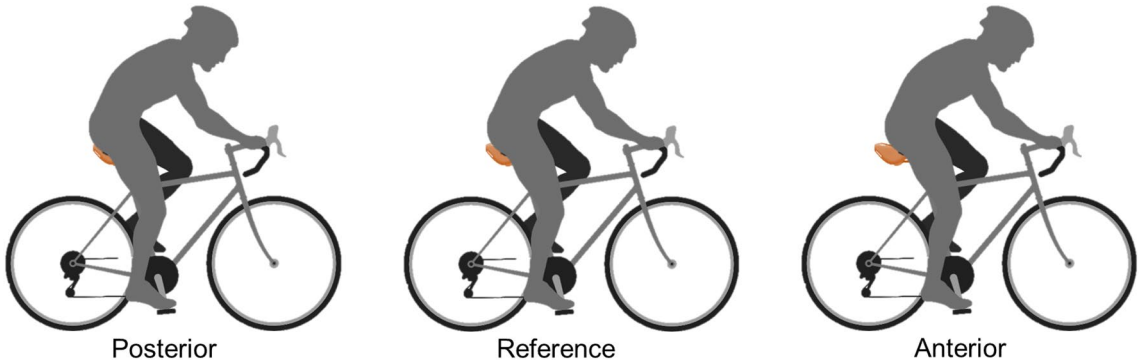


**Fig. 2** Illustration of the three positions on the saddle (posterior, reference, and anterior)

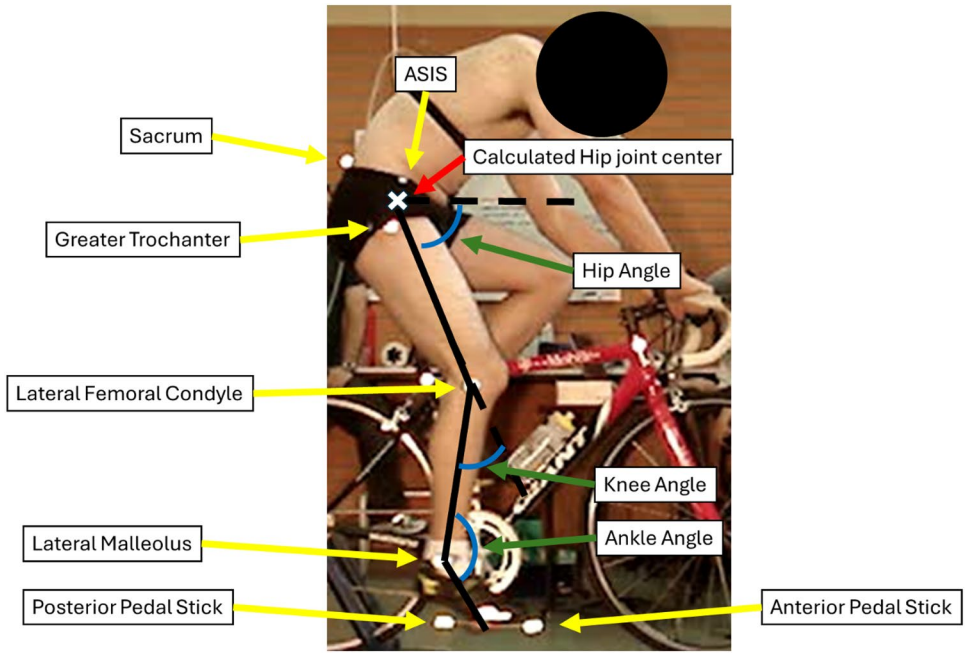


**Fig. 3** Illustration of the landmarks utilised for kinematical analysis and reference frame for angle of the hip, knee and ankle. ASIS stands for anterior superior iliac spine

with a digital second order zero lag low-pass Butterworth recursive filter with cut-off frequency optimized to reduce signal residuals [42]. Optimization of cut-off frequencies started with pre-defined frequencies (i.e. 5 Hz) and trials of ± 30% (at steps of 5%) were conducted to achieve the lowest possible residual (difference between raw data and filtered data). This procedure was conducted for each kinematics waveform. The joint angles of the hip, knee, and ankle during pedalling movement were calculated from the filtered x–y coordinate data, as per the spatial model used by Bini et al. [12] and illustrated in Fig. 3.

Determination of the hip joint centre based on the average coordinate distance between the marker on the anterior superior iliac spine and the greater trochanter was conducted [43]. The average relative horizontal position of the marker on the sacrum to the bottom dead centre was calculated over time across ten pedal revolutions for the analysis of body position on the saddle at the three positions on the saddle (reference, most forward, and most backward).

Three methods were utilised to determine lower limb posture on the bike. The 6 o'clock position method, where angles for the hip, knee, and ankle were extracted when the crank was closest to the bottom dead centre [32]. Hip, knee, and ankle were also extracted at the crank position of the minimum knee flexion angle [i.e. full extension = 0 deg.; 33] and the crank position of the largest distance between the hip joint and the pedal axle [i.e. largest lower limb extension; 34]. These methods are illustrated in Fig. 4. The angular data

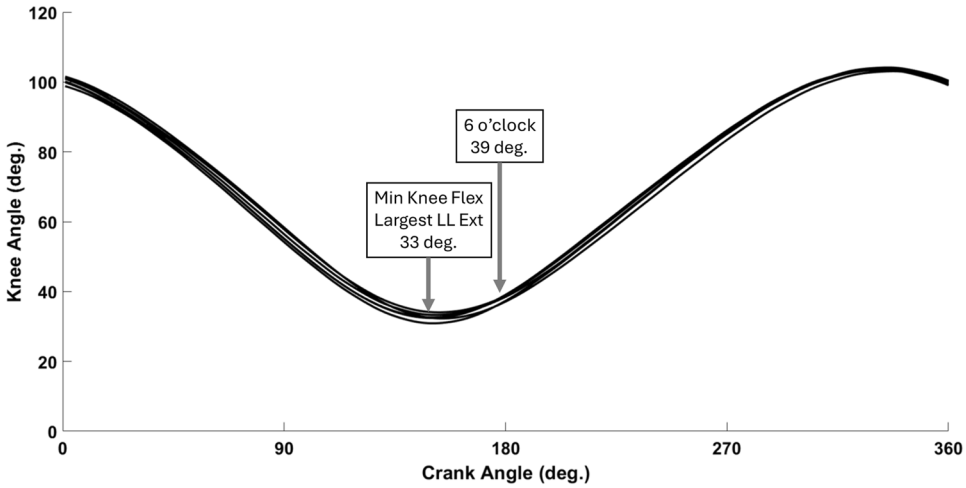


**Fig. 4** Graphical illustration of the 6 o'clock position, minimum knee flexion angle, and largest lower limb (LL) extension methods for the knee angle across the crank cycle. Full extension is 0° of knee flexion

were obtained from 10 consecutive cycles with a mean value per participant for each position taken to statistical analysis.

### Statistical analysis

The normality of data distribution was confirmed for all variables using the Shapiro–Wilk's test. Two-way analysis of variance with repeated measurements was conducted for the hip, knee, and ankle data to assess the main effects from the method of measuring angles (i.e. 6 o'clock position vs. minimum knee angle vs. largest lower limb extension) and from positions on the saddle (i.e. anterior vs. reference vs posterior). Interaction effects were analysed and the assumption of sphericity of the data were confirmed for all variables. Post-hoc analyses were explored when main effects were observed and a correction for multiple comparisons was utilised (i.e. Holm). The effect sizes were also calculated for pairwise comparisons and ranked as trivial ($d < 0.20$), small ($0.20 > d < 0.50$), moderate ($0.50 > d < 0.80$), or large ($d > 0.80$). All analyses were conducted in JASP (Version 0.18.3, University of Amsterdam, The Netherlands) and statistical differences were deemed significant when $p < 0.05$.

## Results

The mean and standard deviation for maximal oxygen uptake, power output, and power output at the second ventilatory threshold were $49.4 \pm 6.6$ ml.kg.min$^{-1}$, $283 \pm 47$ W, and $233 \pm 41$ W, respectively. The most anterior position on the saddle resulted in a $4 \pm 1$ cm change relative to the reference position whilst the most posterior position on the saddle resulted in a change of $3 \pm 1$ cm.

For the hip joint, there were differences between methods but not between positions on the saddle or interaction effects between methods and positions. The hip angle measured at the 6 o'clock position was consistently higher than the use of the minimum knee angle (95% CI = 1.1 to 1.4°, $p < 0.01$, $d = 0.49$–$0.53$—small-moderate) or the largest lower limb extension (95% CI = < 0.2 to 0.5°, $p < 0.01$, $d = 0.13$–$0.20$—trivial). The hip angle measured at the minimum knee flexion was less than when using the largest lower limb extension (95% CI = 0.7 to 1.0°, p < 0.01, d = 0.33–0.38—small).

For the knee joint, the methods and positions differed with an interaction effect observed ($p < 0.01$). When looking at positions on the bike, the knee angle was larger at the anterior position compared to the posterior (95% CI = 8.2 to 13.7°, $p < 0.01$, d = 1.94—large) or the reference positions (95% CI = 2.9 to 8.4°, $p < 0.01$, d = 1.00—large). The reference position presented a larger knee angle compared to the posterior position (95% CI = 2.5 to 8.0°, $p < 0.01$, d = 0.93—large). The knee angle at the 6 o'clock position was larger than the angles at the minimum knee flexion (95% CI = 2.7 to 3.7°, $p < 0.01$, d = 0.57—moderate) or largest lower limb extension (95% CI = 2.7 to 3.6°, $p < 0.01$, d = 0.56—moderate) but no differences were found comparing the minimum knee flexion to the largest lower limb extension ($p = 0.68$).

For the ankle joint, methods and positions differed but no interaction effect was observed. When examining positions on the bike, the ankle angle was less in the most anterior position than the posterior position (95% CI = 3.7 to 8.5°, $p < 0.01$, d = 0.78—moderate), and the reference position (95% CI = 0.4 to 5.2°, $p < 0.01$, $d = 0.37$—small). The reference position also presented a reduced ankle angle from the posterior position (95% CI = 0.8 to 5.6°, $p < 0.01$, $d = 0.41$—small). The ankle angle was larger at the 6 o'clock position than at the minimum knee flexion for the anterior position (95% CI = 0.6 to 1.5°, $p < 0.01$, $d = 0.20$—trivial) and for the reference position (95% CI = 0.5 to 1.4°, $p < 0.01$, $d = 0.17$—trivial). All results are illustrated in Fig. 5.

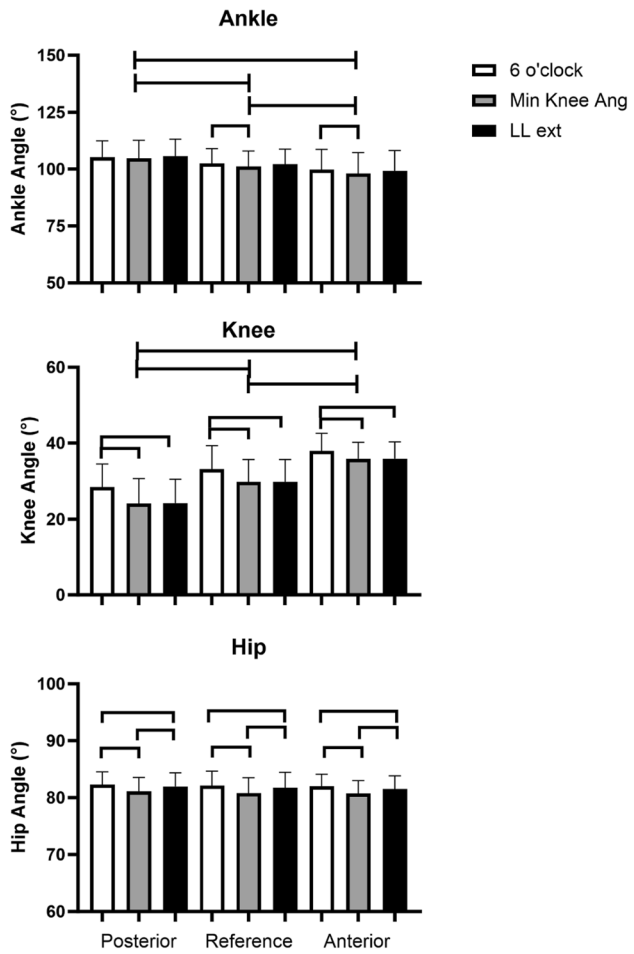


**Fig. 5** Means and standard deviations for ankle, knee, and hip angles obtained at the 6 o'clock position, at minimum knee angle, and at the largest lower limb extension methods of the crank for the posterior, reference, and anterior positions on the saddle. Upper brackets illustrate differences between positions whilst lower brackets illustrate differences between methods

## Discussion and implication

This study illustrated that different methods for determining lower body posture on the bicycle produce different results but provide the ability to detect large changes in position on the saddle. This finding is important because studies have used different methods, which should not be interchanged when aiming for reference angles for posture on the bicycle. As an example, a range of 33–43° of knee flexion has been recommended to determine saddle height [27], which is highly dependent on the method utilised to measure the angle. A second finding was that non-cyclists presented large changes in knee angle when moving forth or back on the saddle, which is in line with data from prior research with cyclists and triathletes [44].

The research on bike fitting and biomechanics of cycling has explored the association between posture on the bike and the likelihood of developing injuries. Holmes et al. [32] proposed an association between too much flexion and anterior knee injuries and excessive extension with hamstring injuries. Data from a systematic review seems to suggest that, the knee flexion angle to optimise performance and reduce the risk of injuries should be between 30 and 43° [31], which is in line with the proposal from Holmes et al. [32]. More recently, studies have attempted to show an association between movement patterns and injuries [20, 21]. These studies utilised reference data from cyclists to attempt setting up desired ranges of motion that could reduce the risk of non-traumatic injuries. Even though the initiative is positive, two barriers need to be resolved. The first is the method to measure angles from a time series of movement data. In the current study, we illustrated that selecting angles from a curve is dependent on when, in the crank cycle, the data is extracted. The second challenge with reference data is the limited research supporting ranges that reduce the risk of injuries. As mentioned before, for determining saddle height, a range of 30–43° of knee flexion has been proposed but no prospective study has been conducted to show that cyclists can truly reduce injuries following this recommendation [31]. In addition, other components of posture on the bike (e.g. upper body posture) have not been explored fully beyond changes in saddle height. Therefore, more research is required to determine a consistent method to describe posture on the bicycle and potential associations between posture and injuries in cycling.

The current research utilised two-dimensional video to calculate joint angles. Even though many studies have used this method, options for movement analysis have been extended recently. The use of gold-standard three-dimensional analysis has been expanding to clinical settings through some commercially available systems (e.g. STT®). Hybrid systems deriving three-dimensional data from a single video have also been used (e.g. Retul®) with good validity compared to traditional three-dimensional systems [45]. Other methods involving the use of inertial measurement units (IMUs) have shown good intra and inter-session reproducibility for cycling [46]. However, other systems have not shown promising validity [47], which could affect the attempt to produce normative data for assessing posture on the bike. Finally, the rapid development of trained neural networks to identify body segments provides a good opportunity for automating the analysis of movement using video [48, 49]. However, future research should be conducted to explore how different systems would affect the angular data obtained from cyclists and, ideally, propose best-practice methods and guidelines for the collection and reporting of kinematic data in cycling.

The rationale for using the 6 o'clock position to determine posture is largely based on the attempt to prevent excessive flexion or extension of the knee, as proposed by Holmes et al. [32]. This method has been later used to describe static posture on the bike [19] but limitations around the type of bicycle should be highlighted. The most important is that bicycles with a shallow seat tube angle or more posterior saddle setback may result in larger differences between the minimum knee flexion and the angle at the 6 o'clock position. This is critical because, the concept of using the 6 o'clock position was largely based on the assumption that, at this point of the crank cycle, the knee would be very close to maximum extension. Therefore, the use of the minimum knee angle may be the most effective way to determine knee flexion, if data from the whole crank cycle is available. This would be feasible when using automated systems to track the lower limb segments. However, when using video to analyse movement, a visual definition of the minimum angle would be challenging without access to the whole knee flexion curve. Moving forward, the use of markerless methods to automatically track segments and produce curves for joint angles should solve this barrier [48, 49], which is already performed by commercial systems designed to extract angles during cycling.

Prior research exploring the implications of movement on the saddle illustrated that cyclists move on the saddle whilst pedalling [50], which is important to acknowledge when considering that posture on the bike is not static. In addition, changes in posture on the bike are observed potentially as an attempt to maximise muscle force production and power output [51]. Taken together, these findings highlight that, for a given bike setup, there are multiple combinations of postures that cyclists could adopt. In the current study, when instructed to move forth or back on the saddle, non-cyclists changed their knee and ankle angles, similar to responses from road cyclists and triathletes [44,

52]. In the current study, no changes were observed for the hip joint, which concurs with a prior study [44], suggesting that more distal joints are subject to changes in movement. Without examining other components such as muscle activity or joint torques, it is difficult to speculate on mechanisms to help explain this response. However, changes in the ankle joint are somewhat expected due to its role in transferring mechanical work from the hip and knee [53]. It is possible that ankle position was changed when moving on the saddle to sustain the same level of force transmission to the pedals. Future research is required though to fully explain this response.

This study was limited to some extent. The use of two-dimensional kinematics is expected to produce ~ 2.2° difference in knee flexion angle compared to three-dimensional data [54]. However, most clinicians would be limited to utilising a single video recording device and data from the current study would be applicable to these scenarios. In addition, measurements of a single limb without bilateral data prevent the analysis of movement asymmetries. Healthy male non-cyclists were recruited to avoid adaptations seen in trained cyclists to a given position on the saddle. This approach limits the application of these findings to a female population. Some data suggest that females opt for a different posture on the bike [55], which warrants new research in this population.

## Conclusion

Determining lower limb position on the bike using different methods produces different data but the three methods enable the detection of large changes in posture on the saddle. Moving forward or backward on the saddle leads to large changes in knee angle and small to moderate changes in the ankle with no changes at the hip. The angles obtained at the 6 o'clock position, minimum knee angle, or largest lower limb extension should not be used interchangeably.

**Supplementary Information** The online version contains supplementary material available at https://doi.org/10.1007/s11332-025-01326-9.

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**Data availability** Data is provided within the manuscript or supplementary information files.

## Declarations

**Conflict of interest** The authors declare that there is no conflict of interest with the content of this research.

**Ethical approval** This study was approved by the University Human Ethics Committee (Project number 17684), according to the Declaration of Helsinki.

**Informed consent** Before data collection, all cyclists signed an informed consent to participate in the study.

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